TUTORIAL V

TABLE OF CONTENTS, INDEX AND GLOSSARY

V.1. TABLE OF CONTENTS

A *table of contents* is a special list which contains the section numbers and corresponding headings as given in the standard form of the sectioning commands, together with the page numbers on which they begin. Similar lists exist containing reference information about the floating elements in a document, namely, the *list of tables* and *list of figures*. The structure of these lists is simpler, since their contents, the captions of the floating elements, all are on the same level.

Standard LaTeX can automatically create these three contents lists. By default, LaTeX enters text generated by one of the arguments of the sectioning commands into the `.toc` file. Similarly, LaTeX maintains two more files, one for the list of figures (`.lof`) and one for the list of tables (`.lot`), which contain the text specified as the argument of the `\caption` command for figures and tables.

`\tableofcontents` produces a table of contents. `\listoffigures` and `\listoftables` produce a list of figures and list of tables respectively. These lists are printed at the point where these commands are issued. Occasionally, you may find that you do not like the way LaTeX prints a table of contents or a list of figures or tables. You can fine-tune an individual entry by using the optional arguments to the sectioning command or `\caption` command that generates it. Formatting commands can also be introduced with the `\addtocontents`. If all else fails, you can edit the `.toc`, `lof`, `lot` files yourself. Edit these files only when preparing the final version of your document, and use a `\nofiles` command to suppress the writing of new versions of the files.

V.1.1. Additional entries

The *-form sectioning commands are not entered automatically in the table of contents. LaTeX offers two commands to insert such information directly into a contents file:

`\addtocontents{`*file*`}{`*text*`}` `\addcontentsline{`*file*`}{`*type*`}{`*text*`}`

file	The extension of the contents file, usually `toc`, `lof` or `lot`.
type	The type of the entry. For the `toc` file the *type* is normally the same as the heading according to the format of which an entry must be typeset. For the `lof` or `lot` files, `figure` or `table` is specified.
text	The actual information to be written to the *file* mentioned. LaTeX commands should be protected by `\protect` to delay expansion

The `\addtocontents` command does not contain a *type* parameter and is intended to enter *user-specific* formatting information. For example, if you want to generate additional spacing in the middle of a table of contents, the following command can be issued:

```
\addtocontents{toc}{\protect\vspace{2ex}}
```

The \addcontentsline instruction is usually invoked *automatically* by the document sectioning commands, or by the \caption commands. If the entry contains numbered text, then \numberline must be used to separate the section number (*number*) from the rest of the text for the entry (*heading*) in the *text* parameter:

\protect\numberline{*number*}{*heading*}

For example, a \caption command inside a figure environment saves the text annotating the figure as follows:

\addcontentsline{lof}{figure}{\protect\numberline{\thefigure}*captioned text*}

Sometimes \addcontentsline is used in the source to complement the actions of standard LaTeX. For instance, in the case of the starred form of the section commands, no information is written to the .toc file. So if you do not want a heading number (starred form) but an entry in the .toc file you can write something like:

```
\chapter*{Forward}
\addcontentsline{toc}{chapter}{\numberline{}Forward}
```

This produces an indented "chapter" entry in the table of contents, leaving the space where the chapter number would go free. Omitting the \numberline command would typeset the word "Forward" flush left instead.

V.1.2. Typesetting a contents list

As discussed above, contents lists consist of entries of different types, corresponding to the structural units that they represent. Apart from these standard entries, these lists may contain any commands. A standard entry is specified by the command:

\contentsline{*type*}{*text*}{*page*}

type Type of the entry, e.g. section, or figure.
text Actual text as specified in the argument of the sectioning or \caption commands.
page Pagenumber.

Note that section numbers are entered as a parameter of the \numberline command to allow formatting with the proper indentation. It is also possible for the user to create a table of contents by hand with the help of the command \contentsline. For example:

```
\contentsline {section}
  {\numberline {2.4}Structure of the Table of Contents}{31}
```

To format an entry in the table of contents files, standard LaTeX makes use of the following command:

\@dottedtocline{*level*}{*indent*}{*numwidth*}{*text*}{*page*}

The last two parameters coincide with those of \contentsline, since the latter usually invokes \@dottedtocline command. The other parameters are the following:

level The nesting level of an entry. This parameter allows the user to control how many nesting levels will be displayed. Levels greater than the value of counter tocdepth will not appear in the table of contents.
indent This is total indentation from the left margin.
numwidth The width of the box that contains the number if *text* has a \numberline command. This is also the amount of extra indentation added to the second and later lines of a multiple line entry.

Additionally, the command \@dottedtocline uses the following formatting parameters, which specify the visual appearance of all entries:

\@pnumwidth	The width of the box in which the page number is set.
\@tocmarg	The indentation of the right margin for all but the last line of multiple line entries. Dimension, but changed with \renewcommand.
\@dotsep	The separation between dots, in mu (math units). It is a pure number (like 1.7 or 2). By making this number large enough you can get rid of the dots altogether. Changed with \renewcommand as well.

V.1.3. Multiple tables of contents

The minitoc package, initially written by Nigel Ward and Dan Jurafsky and completely redesigned by Jean-Pierre Drucbert, creates a mini-table of contents (a "minitoc") at the beginning of each chapter when you use the book or report classes.

The mini-table of contents will appear at the beginning of a chapter, after the \chapter command. The parameters that govern the use of this package are discussed below:

Table V.1: Summary of the minitoc parameters

\dominitoc	Must be put just in front of \tableofcontents, to initialize the minitoc system (Mandatory).
\faketableofcontents	This command replaces \tableofcontents when you want minitocs but not table of contents.
\minitoc	This command must be put right after each \chapter command where a minitoc is desired.
\minitocdepth	A LaTeX counter that indicates how many levels of headings will be displayed in the minitoc (default value is 2).
\mtcindent	The length of the left/right indentation of the minitoc (default value is 24pt).
\mtcfont	Command defining the font that is used for the minitoc entries (The default definition is a small roman font).

For each mini-table, an auxiliary file with extension .mtc<N> where <N> is the chapter number, will be created.

By default, these mini-tables contain only references to sections and subsections. The minitocdepth counter, similar to tocdepth, allows the user to modify this behaviour.

As the minitoc takes up room on the first page(s) of a chapter, it will alter the page numbering. Therefore, three runs normally are needed to get correct information in the mini-table of contents.

To turn off the \minitoc commands, merely replace the package minitoc with minitocoff on your \usepackage command. This assures that all \minitoc commands will be ignored.

V.2. INDEX

To find a topic of interest in a large document, book, or reference work, you usually turn to the table of contents or, more often, to the index. Therefore, an index is a very important part of a document, and most users' entry point to a source of information is precisely through a pointer in the index. The most generally used index preparation program is *MakeIndex*.

```
Page vi:  \index{animal}
Page 5:   \index{animal}
Page 6:   \index{animal}
Page 7:   \index{animal}
Page 11: \index{animalism|see{animal}}
Page 17: \index{animal@\emph{animal}}
         \index{mammal|textbf}
Page 26: \index{animal!mammal!cat}
Page 32: \index{animal!insect}
```

(a) The input file

```
\indexentry{animal}{vi}
\indexentry{animal}{5}
\indexentry{animal}{6}
\indexentry{animal}{7}
\indexentry{animalism|seeanimal}{11}
\indexentry{animal@\emph{animal}}{17}
\indexentry{mammal|textbf}{17}
\indexentry{animal!mammal!cat}{26}
\indexentry{animal!insect}{32}
```

(b) The `.idx` file

```
\begin{theindex}
  \item animal, vi, 5–7
    \subitem insect, 32
    \subitem mammal
      \subsubitem cat, 26
  \item \emph{animal}, 17
  \item animalism, \see{animal}{11}
\indexspace
  \item mammal, \textbf{17}
\end{theindex}
```

(c) The `.ind` file

animal, vi 5–7
    insect, 32
    mammal
        cat, 26
animal, 17
animalism, *see* animal

mammal, **17**

(d) The typeset output

Figure V.1: Stepwise development of index processing

Each `\index` command causes LaTeX to write an entry in the `.idx` file. This command writes the text given as an argument, in the `.idx` file. This `.idx` will be generated only if we give `\makeindex` command in the preamble otherwise it will produce nothing.

`\index{`*index_entry*`}`

To generate index follow the procedure given below:

1. Tag the words inside the document, which needs to come as index, as an argument of `\index` command.
2. Include the makeidx package with an `\usepackage` command and put `\makeindex` command at the preamble.
3. Put a `\printindex` command where the index is to appear, normally before `\end{document}` command.
4. LaTeX *file*. Then a raw index (`file.idx`) will be generated.
5. Then run `makeindex`. (makeindex *file.idx* or makeindex *file*). Then two more files will be generated, *file.ind* which contains the index entries and *file.ilg*, a transcript file.
6. Then run LaTeX again. Now you can see in the dvi that the index has been generated in a new page.

V.2.1. Simple index entries

Each `\index` command causes LaTeX to write an entry in the `.idx` file. For example

`\index{`*index_entry*`}`

```
fonts                                  Page ii:  \index{table|(}
      Computer Modern, 13–25           Page xi:  \index{table|)}
      math, see math, fonts            Page 5:   \index{fonts!PostScript|(}
      PostScript, 5                              \index{fonts!PostScript|)}
table, ii–xi, 14                       Page 13   \index{fonts!Computer Modern |(}
                                       Page 14: \index{table}
                                       Page 17: \index{fonts!math|see{math, fonts}}
                                       Page 21: \index{fonts!Computer Modern}
                                       Page 25: \index{fonts!Computer Modern|)}
```

Figure V.2: Page range and cross-referencing

V.2.2. Sub entries

Up to three levels of index entries (main, sub, and subsub entries) are available with LaTeX-`MakeIndex`. To produce such entries, the argument of the `\index` command should contain both the main and subentries, separated by `!` character.

Page 5: `\index{`*dimensions!rule!width*`}`

This will come out as

dimensions
    rule
        width, 5

V.2.3. Page ranges and cross-references

You can specify a page range by putting the command `\index{...|(}` at the beginning of the range and `\index{...|)}` at the end of the range. Page ranges should span a homogeneous numbering scheme (e.g., Roman and Arabic page numbers cannot fall within the same range).

You can also generate cross-reference index entries without page numbers by using the `see` encapsulator. Since "see" entry does not print any page number, the commands `\index{...|see{...}}` can be placed anywhere in the input file after the `\begin{document}` command. For practical reasons, it is convenient to group all such cross-referencing commands in one place.

V.2.4. Controlling the presentation form

Sometimes you may want to sort an entry according to a key, while using a different visual representation for the typesetting, such as Greek letters, mathematical symbols, or specific typographic forms. This function is available with the syntax: *key@visual*, where *key* determines the alphabetical position and the string *value* produces the typeset text of the entry.

For some indexes certain page numbers should be formatted specially, with an italic page number (for example) indicating a primary reference, and an *n* after a page number denoting that the item appears in a footnote on that page. *MakeIndex* allows you to format an individual page number in any way you want by using the encapsulator syntax specified | character. What follows the | sign will "encapsulate" or enclose the page number associated with the index entry. For instance, the command `\index{keyword|xxx}` will produce a page number of the form `\xxx{n}`, where *n* is the page number in question.